

Experiment 1: Installing Oracle / MySQL

Aim: Installation of MySQL

Software Required:

- Windows OS
- Internet Connection
- MySQL Workbench Installer

Procedure:

1. Open browser and go to the official MySQL website:
<https://dev.mysql.com/downloads/workbench/>
2. Click on **Download Now** button.



3. Click on **Download** button for Windows (MSI Installer).
4. Click on “No thanks, just start my download.”

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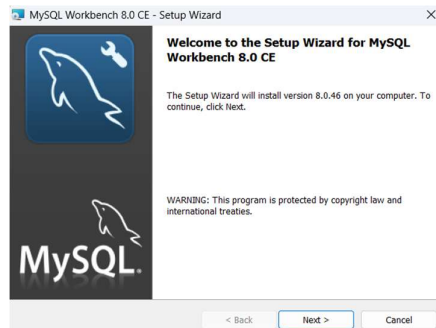
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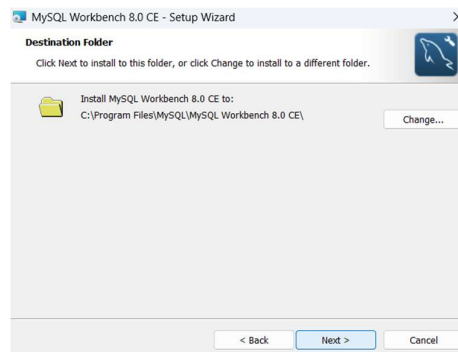
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5. After download completes, double-click the .msi file.

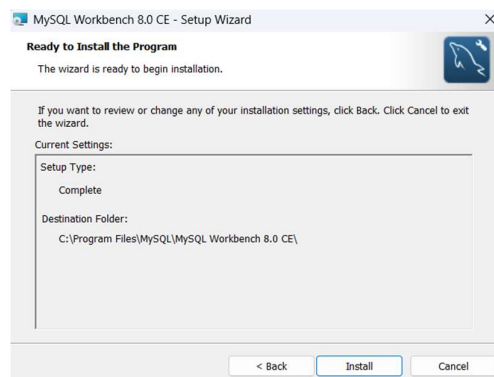
6. Setup Wizard opens → Click **Next**.



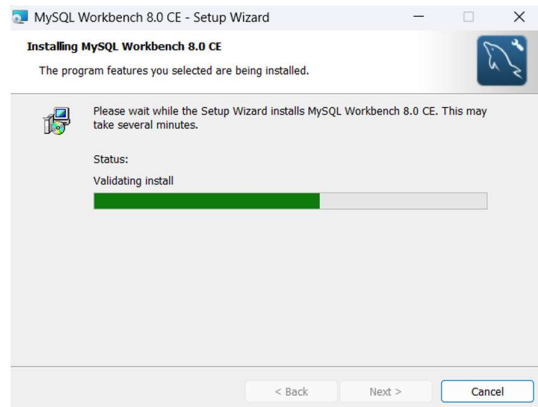
7. Select **Complete Setup Type** → Click **Next**.



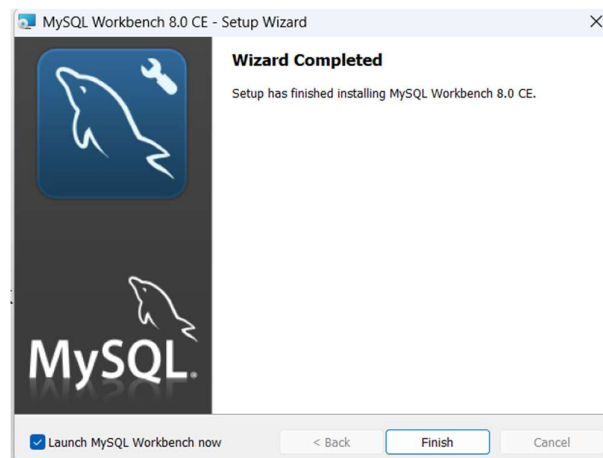
8. Click **Install**.



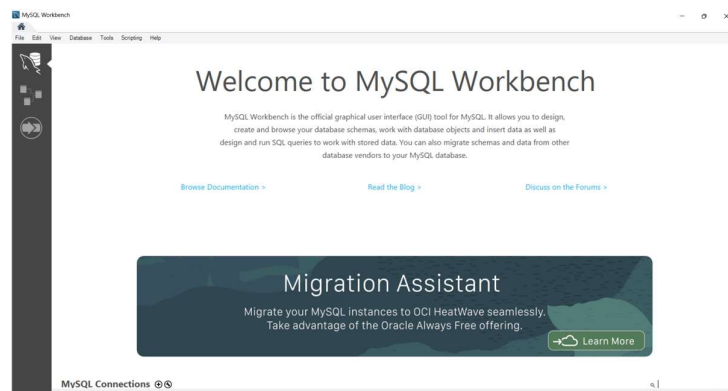
9. Wait until installation completes.



10. Click **Finish**.



11. MySQL Workbench opens successfully.



Result : MySQL Workbench is installed successfully on the system.

Experiment 2: DDL Commands (Create, Alter, Drop)

Theory: DDL (Data Definition Language) commands are used to define database structure. Common

Commands: CREATE – creates tables ALTER – modifies tables DROP – deletes tables SQL

Example: CREATE TABLE student (roll_no INT PRIMARY KEY, name VARCHAR(20),age INT, subject VARCHAR(30));

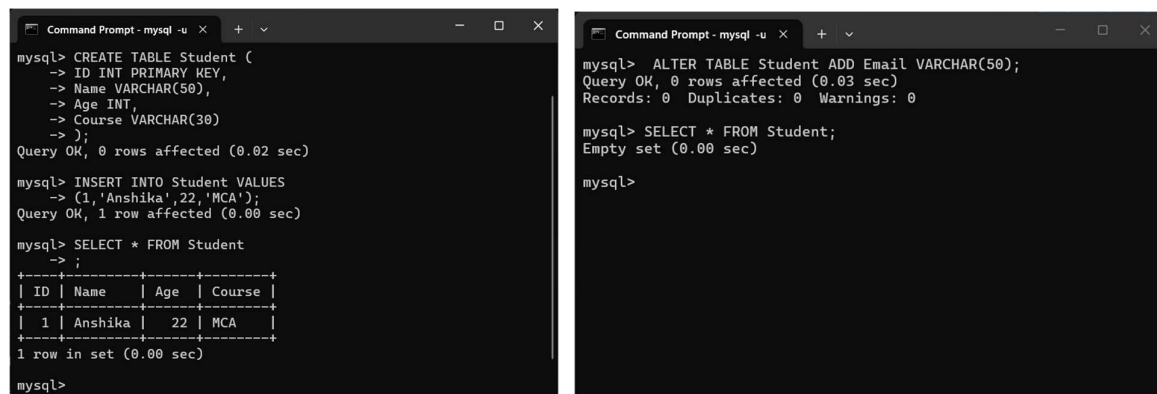
ALTER TABLE student ADD address

VARCHAR(50); DROP TABLE student;

Procedure :

- | | | |
|------------------|-------------------|------------------------|
| 1- CREATE TABLE | 2- ALTER TABLE | 3- DROP TABLE Student; |
| Student (ID INT | Student ADD Email | |
| PRIMARY KEY, | VARCHAR(50); | |
| Name | ALTER TABLE | |
| VARCHAR(50), | Student | |
| Age INT, | MODIFY Age | |
| Course | INT; | |
| VARCHAR(30) | | |
|); | | |

Output Screenshot



```
mysql> CREATE TABLE Student (
  -> ID INT PRIMARY KEY,
  -> Name VARCHAR(50),
  -> Age INT,
  -> Course VARCHAR(30)
  -> );
Query OK, 0 rows affected (0.02 sec)

mysql> INSERT INTO Student VALUES
  -> (1,'Anshika',22,'MCA');
Query OK, 1 row affected (0.00 sec)

mysql> SELECT * FROM Student
  -> ;
+----+-----+-----+-----+
| ID | Name  | Age  | Course |
+----+-----+-----+-----+
| 1  | Anshika | 22  | MCA    |
+----+-----+-----+-----+
1 row in set (0.00 sec)

mysql>

mysql> ALTER TABLE Student ADD Email VARCHAR(50);
Query OK, 0 rows affected (0.03 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Student;
Empty set (0.00 sec)

mysql>
```

```
Command Prompt - mysql -u x + v - □ ×
mysql> DROP TABLE Student;
Query OK, 0 rows affected (0.02 sec)

mysql>
```

Experiment 3: Entity Relationship Diagram

Theory: An ER diagram represents entities, attributes, and relationships in a database.

Steps:

1. Identify entities (Student, Course, Teacher).
2. Identify attributes (Name, ID, Age).
3. Define relationships between entities.
4. Use tools like MySQL Workbench or Draw.io.

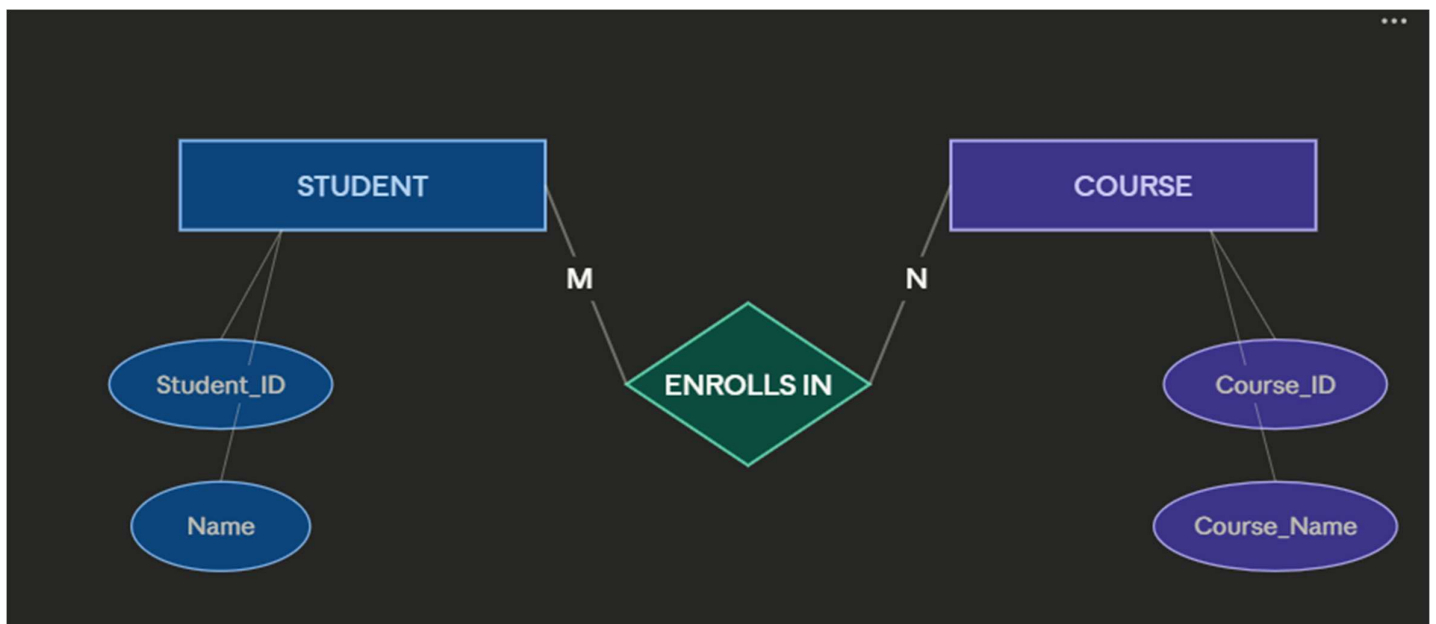
Example: Entities: Student (Student_ID, Name) Course (Course_ID, Course_Name)

Procedure

Student ----- Enrolls ----- Course

Student_ID (PK)	Course_ID (PK)
Name	Course_Name
Age	Duration

Output Screenshot :



Experiment 4: DDL and DML Commands

Theory: DML commands manipulate data inside tables.

Commands:

- INSERT – add records
- UPDATE – modify records
- DELETE – remove records

SQL Example:

- INSERT INTO student VALUES(1,'Anshika',21,'DBMS');
- UPDATE student SET age=23 WHERE roll_no=1;
- DELETE FROM student WHERE roll_no=1;

Procedure

- | | | |
|---|---|---|
| 1- CREATE TABLE Student (
Student_ID INT PRIMARY
KEY, Name VARCHAR(50)
NOT NULL, Age INT
CHECK (Age >= 18),
Course VARCHAR(30)
DEFAULT 'BCA', Email
VARCHAR(50) UNIQUE
); | 2- INSERT INTO Student VALUES
(101, 'Rahul', 20, 'BCA',
'rahul@gmail.com');

INSERT INTO Student VALUES
(102, 'Amit', 19, 'BCA',
'amit@gmail.com');

INSERT INTO Student VALUES
(103, 'Priya', 21, 'BBA',
'priya@gmail.com'); | 3- UPDATE
Student SET
Course =
'B.Tech'
WHERE Student_ID = 102; |
|---|---|---|
- 4- DELETE FROM Student WHERE Student_ID = 103;

Output Screenshot

```
Command Prompt - mysql -u x + v
mysql> CREATE TABLE Student ( Student_ID INT PRIMARY KEY, Name V
ARCHAR(50) NOT NULL, Age INT CHECK (Age >= 18),Course VARCHAR(30
) DEFAULT 'MCA', Email VARCHAR(50) UNIQUE);
Query OK, 0 rows affected (0.07 sec)

mysql> INSERT INTO Student (Student_ID, Name, Age, Course, Email
)
-> VALUES (1, 'Anshika', 20, 'BCA', 'anshika@example.com');
Query OK, 1 row affected (0.04 sec)

mysql> SELECT * FROM Student
-> ;
+-----+-----+-----+-----+-----+
| Student_ID | Name   | Age | Course | Email               |
+-----+-----+-----+-----+-----+
| 1          | Anshika | 20  | BCA    | anshika@example.com |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql>
```

```
Command Prompt - mysql -u x + v
mysql> UPDATE Student SET Course = 'MCA' WHERE Student_ID = 1;
Query OK, 1 row affected (0.04 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT * FROM Student
-> ;
+-----+-----+-----+-----+-----+
| Student_ID | Name   | Age | Course | Email               |
+-----+-----+-----+-----+-----+
| 1          | Anshika | 20  | MCA    | anshika@example.com |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql>
```

```
Command Prompt - mysql -u x + v
mysql> SELECT * FROM Student;
+-----+-----+-----+-----+-----+
| Student_ID | Name   | Age | Course | Email               |
+-----+-----+-----+-----+-----+
| 1          | Anshika | 20  | MCA    | anshika@example.com |
| 2          | Shivansh | 21  | BCA    | shivansh@example.com |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> DELETE FROM Student WHERE Student_ID = 1;
Query OK, 1 row affected (0.04 sec)

mysql> SELECT * FROM Student;
+-----+-----+-----+-----+-----+
| Student_ID | Name   | Age | Course | Email               |
+-----+-----+-----+-----+-----+
| 2          | Shivansh | 21  | BCA    | shivansh@example.com |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql>
```


Experiment 5: Basic SELECT Statements

Theory: SELECT statements retrieve data from tables.

Examples:

- SELECT * FROM student;
- SELECT name FROM student;
- SELECT * FROM student WHERE age > 20;
- SELECT * FROM student ORDER BY name;

Procedure:

SYNTAX

SELECT
column_name FROM
table_name;

EXAMPLE

SELECT * FROM Student;

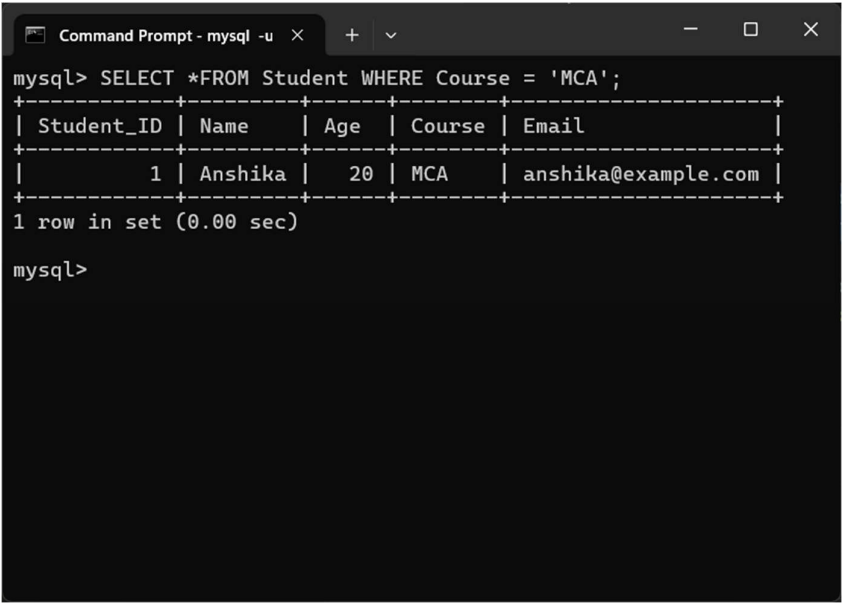
Restricting Data (Using WHERE Clause):

The WHERE clause is used to filter records based on a condition.

Procedure:

EXAMPLE: Display students whose course is MCA: SELECT *FROM Student WHERE Course = 'MCA';

Output Screenshot:



```
Command Prompt - mysql -u × + ∨
mysql> SELECT *FROM Student WHERE Course = 'MCA';
+-----+-----+-----+-----+-----+
| Student_ID | Name   | Age  | Course | Email                      |
+-----+-----+-----+-----+-----+
| 1          | Anshika | 20   | MCA    | anshika@example.com       |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql>
```

Experiment 6: Aggregate Functions

Theory: Aggregate functions perform calculations on multiple rows.

Functions: COUNT() SUM() AVG() MAX() MIN()

Example:

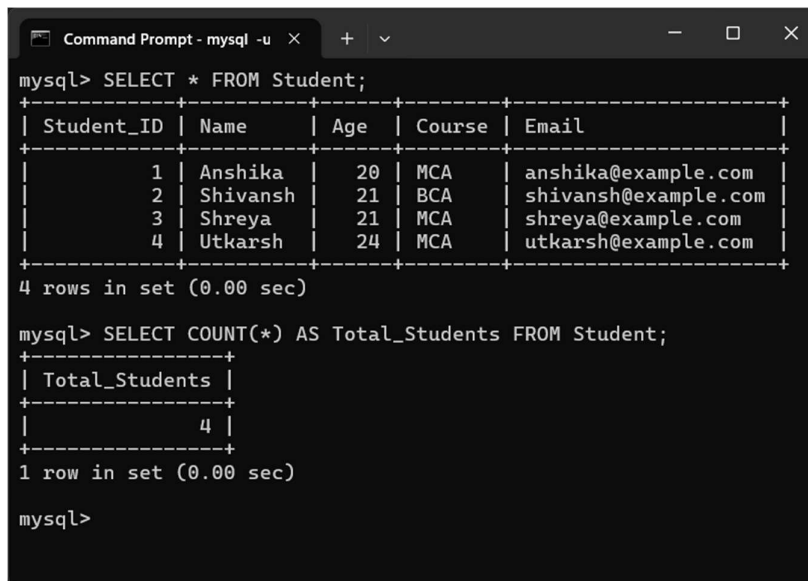
- SELECT COUNT(*) FROM student;
- SELECT AVG(age) FROM student;

Procedure:

- Open MySQL / Oracle database software.
- Create a table using the CREATE TABLE command.
- Insert records into the table using the INSERT INTO statement.
- Retrieve data from the table using the SELECT command.
- Use aggregate functions such as COUNT(), SUM(), AVG(), MAX(), and MIN() to perform calculations on the data.
- Use the LIKE operator with wildcards (%) and _ for character matching to retrieve specific records.
- Execute the SQL queries and verify the output.

Output Screenshot:

- Aggregate Function
 - Count number of students:



```
mysql> SELECT * FROM Student;
+-----+-----+-----+-----+-----+
| Student_ID | Name   | Age | Course | Email                |
+-----+-----+-----+-----+-----+
| 1          | Anshika | 20  | MCA    | anshika@example.com  |
| 2          | Shivansh | 21  | BCA    | shivansh@example.com |
| 3          | Shreya  | 21  | MCA    | shreya@example.com   |
| 4          | Utkarsh | 24  | MCA    | utkarsh@example.com  |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> SELECT COUNT(*) AS Total_Students FROM Student;
+-----+
| Total_Students |
+-----+
| 4              |
+-----+
1 row in set (0.00 sec)

mysql>
```

- Retrieval by character matching

Names starting with S

```
Command Prompt - mysql -u x + v - □ X
mysql> SELECT * FROM Student WHERE Name LIKE 'S%';
+-----+-----+-----+-----+-----+
| Student_ID | Name      | Age | Course | Email                      |
+-----+-----+-----+-----+-----+
|          2 | Shivansh  | 21  | BCA    | shivansh@example.com      |
|          3 | Shreya    | 21  | MCA    | shreya@example.com        |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql>
```

Experiment 7: SQL Clauses

Theory: Clauses filter and organize query results.

Examples:

- `SELECT * FROM student ORDER BY name;`
- `SELECT subject, COUNT(*) FROM student GROUP BY subject;`
- `SELECT * FROM student WHERE age BETWEEN 18 AND 25;`
- `SELECT * FROM student WHERE subject IN ('DBMS','OS');`

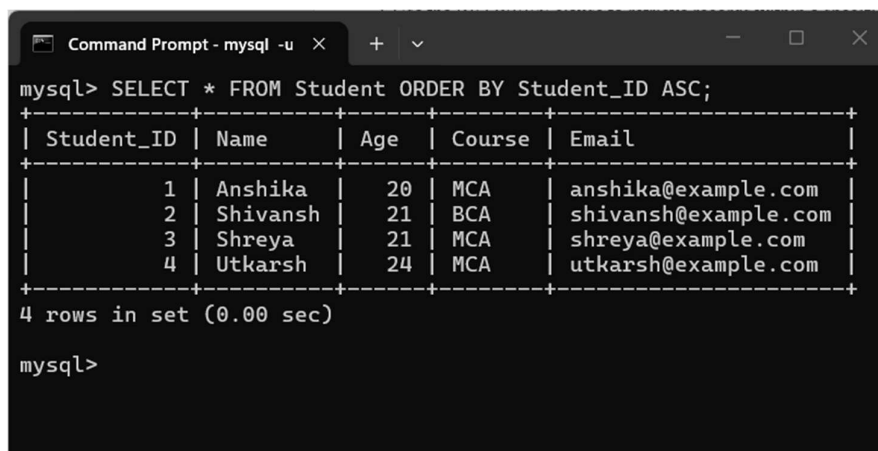
Procedure

1. Open MySQL / Oracle database environment.
2. Create a table using the `CREATE TABLE` command.
3. Insert records into the table using the `INSERT INTO` statement.
4. Use the `ORDER BY` clause with the `SELECT` statement to sort records in ascending or descending order.
5. Use the `GROUP BY` clause to group rows having similar values and apply aggregate functions like `COUNT()` or `AVG()`.
6. Use the `IN` clause in the `WHERE` condition to retrieve records that match multiple values.
7. Use the `BETWEEN` clause to retrieve records within a specified range.
8. Execute the SQL queries and observe the output.

Output Screenshot

1. ORDER BY Clause

Sort students by Age in ascending order:



```
Command Prompt - mysql -u x + v
mysql> SELECT * FROM Student ORDER BY Student_ID ASC;
+-----+-----+-----+-----+-----+
| Student_ID | Name   | Age | Course | Email                |
+-----+-----+-----+-----+-----+
| 1          | Anshika | 20  | MCA    | anshika@example.com  |
| 2          | Shivansh | 21  | BCA    | shivansh@example.com |
| 3          | Shreya  | 21  | MCA    | shreya@example.com   |
| 4          | Utkarsh | 24  | MCA    | utkarsh@example.com  |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql>
```

GROUP BY Clause

Count number of students in each course:

```
Command Prompt - mysql -u x + v
mysql> SELECT Course, COUNT(*) FROM Student GROUP BY Course;
+-----+-----+
| Course | COUNT(*) |
+-----+-----+
| MCA    | 3        |
| BCA    | 1        |
+-----+-----+
2 rows in set (0.00 sec)

mysql> |
```

IN Clause

Display students who belong to BCA or BBA course.

```
Command Prompt - mysql -u x + v
mysql> SELECT * FROM Student WHERE Course IN ('BCA','MCA');
+-----+-----+-----+-----+-----+
| Student_ID | Name    | Age | Course | Email                |
+-----+-----+-----+-----+-----+
| 1          | Anshika | 20  | MCA    | anshika@example.com  |
| 2          | Shivansh | 21  | BCA    | shivansh@example.com |
| 3          | Shreya  | 21  | MCA    | shreya@example.com   |
| 4          | Utkarsh | 24  | MCA    | utkarsh@example.com  |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> |
```

BETWEEN Clause

Display students whose Age are between 20 and 25.

```
Command Prompt - mysql -u x + v
mysql> SELECT * FROM Student WHERE Age BETWEEN 20 AND 25;
+-----+-----+-----+-----+-----+
| Student_ID | Name   | Age | Course | Email                |
+-----+-----+-----+-----+-----+
| 1 | Anshika | 20 | MCA    | anshika@example.com |
| 2 | Shivansh | 21 | BCA    | shivansh@example.com |
| 3 | Shreya  | 21 | MCA    | shreya@example.com   |
| 4 | Utkarsh | 24 | MCA    | utkarsh@example.com  |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> |
```